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Question	Answer	Marks
1	**Defining the problem**	
	m is the independent variable and *v* is the dependent variable **or** vary *m* and measure *v*	**1**
	keep *P* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • *P* attached to string • string passing over a supported pulley • string horizontally attached to trolley • labels for pulley and *P*	**1**
	method to <u>determine *v*</u> e.g. • light gate attached to a timer to <u>measure *v*</u> • motion sensor connected to data logger • ticker tape timer • measure *d*, and *t* with timer/stopwatch/light gate and timer (to calculate *v*)	**1**
	use a (top pan) balance to measure *m*	**1**
	measure *d* with a ruler	**1**
	Method of Analysis	
	plot a graph of $1 / v^2$ against *m* **or** *m* against $1 / v^2$	**1**
	$Q = Pg - \frac{1}{2d \times \text{gradient}}$ **or** $Q = Pg - \frac{\text{gradient}}{2d}$ (consistent with graph)	**1**
	$R = 2d \times (Pg - Q) \times y\text{-intercept}$ $(= \frac{y\text{-intercept}}{\text{gradient}})$ **or** $R = -y\text{-intercept}$ (consistent with graph)	**1**

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1	**Additional detail including safety considerations**		**6**
	D1	safety precaution related to falling mass *P* or preventing trolley falling e.g. cushion/sandbox for *P*, barrier for trolley	
	D2	keep *d* constant	
	D3	use a large distance for *d*	
	D4	method to keep *d* constant, e.g. mark starting and end positions	
	D5	*v* = distance / time for appropriate small distance on ticker tape/card/distance between light gates (**not** *d*) **or** use of $v = 2d / t$ if *d* and total time measured	
	D6	method to ensure wooden surface is horizontal, e.g. spirit level	
	D7	for same *m* repeat experiment to find average *v* or *t*	
	D8	$\frac{1}{v^2} = \frac{m}{2d \times (Pg - Q)} + \frac{R}{2d \times (Pg - Q)}$ **or** $m = \frac{2d(Pg - Q)}{v^2} - R$ (consistent with graph)	
	D9	relationship valid if a straight line (not passing through the origin)	
	D10	method of attaching mass *m* to trolley e.g. tape, adhesive putty	